COME: TEST-TIME ADAPTION BY CONSERVATIVELY MINIMIZING ENTROPY

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Overconfidence makes model collapse

- Entropy Minimization can be highly unstable and frequently lead to model collapse when the models encounter unreliable samples in the wild
  - Samples selection
  - Constrained optimization
  - Model recovery

➢ Propose Conservatively Minimizing Entropy (COME)
  - Uses Subjective Logic
  - Decides whether to trust or not by a data-adaptive upper bound on the model confidence

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Uncertainty quantification

- Quantitatively characterize the probability that the model prediction is correct

- To obtain the uncertainty
  - Bayesian neural networks (BNNs): replace the deterministic weight parameters of model with distribution
  - Ensemble-based methods: train multiple models and ensemble them
  - Subjective Logic (SL)

- COME uses SL
  - Models the uncertainty in a single forward pass without modifying the training strategy or model architecture

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The overconfident issue of EM

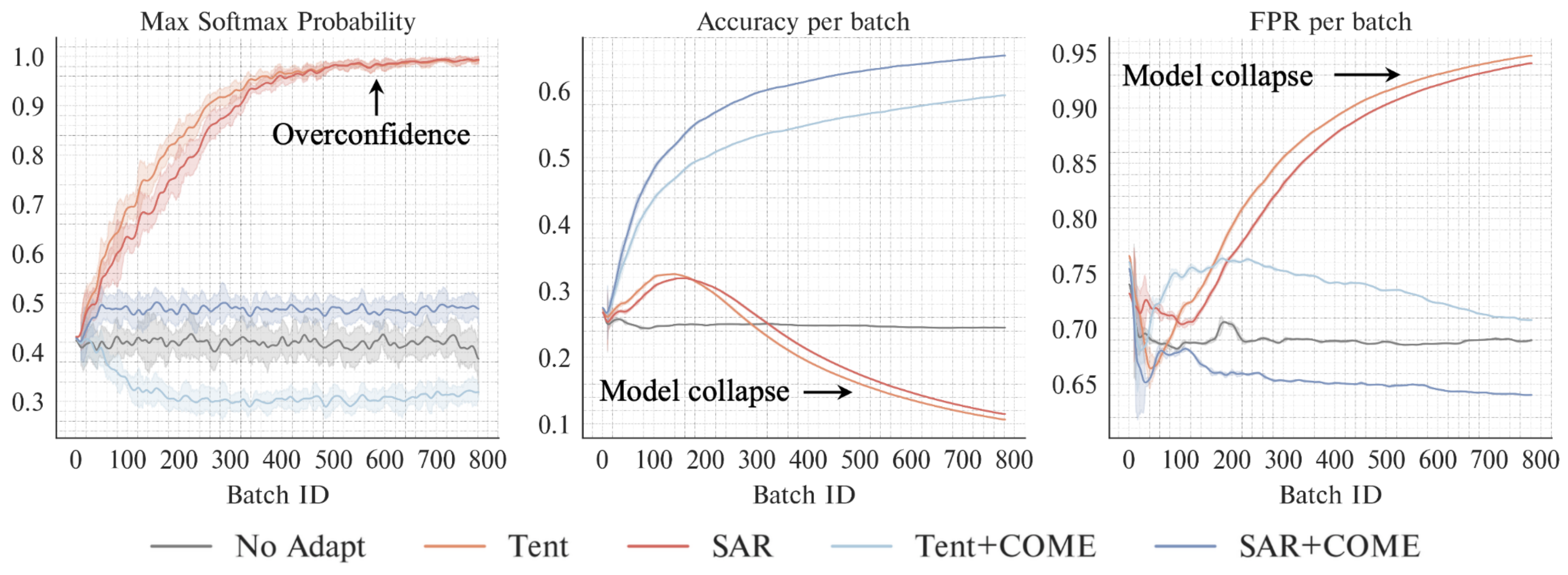

Figure 1: Empirical observations of Entropy Minimization when equipped to two representative TTA methods, i.e., the seminal Tent (Wang et al., 2021) and recent advanced SAR (Niu et al., 2023). Along the TTA process, the uncertainty of models tuned with EM quickly drops, and the false positive rate decreases temporarily for a very short time horizon before quickly increasing. Along the same adaption trajectory, the model accuracy also improves for a short time compared to the initial model and then quickly decreases, after which the model collapses to a trivial solution. By contrast to EM, our COME establishes a stable TTA process with consistently improved classification accuracy and false positive rate. Besides, the model confidence of our COME is much more conservative, which implies fewer risks of overconfidence and a more accurate uncertainty awareness.

The overconfident issue of EM

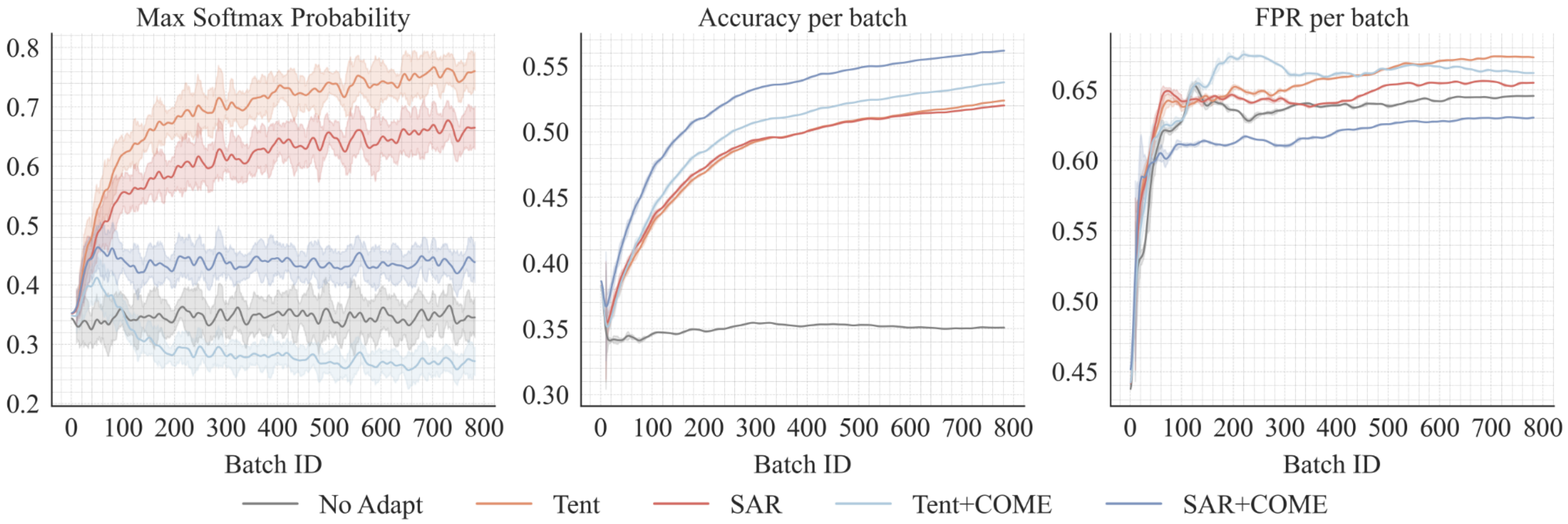

Figure 3: Comparison on two representative TTA methods on ImageNet-C under **Gaussian Noise** corruption of severity level 5. By contrast to EM, our `COME` establishes a stable TTA process with consistently improved classification accuracy and false positive rate.

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Subjective Logic

- Softmax probability often leads to overconfident predictions, even when the predictions are wrong or the inputs are abnormal outliers

➢ Propose to obtain the uncertainty through subjective logic

Subjective Logic

- Softmax function that directly normalizes the logits f (x) to model the class distribution p(y|x)

- SL considers the model output as evidence (denoted as e) to model a Dirichlet distribution which represents the density of all possible probability assignment μ = [p(y = 1|x), p(y = 2|x), . . . , p(y = K|x)]

- Dirichlet distribution

$$p(\boldsymbol{\mu}|x) = \mathrm{Dir}(\boldsymbol{\mu}|\boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})}\prod_{k=1}^{K}\mu_k^{\alpha_k - 1}, \quad \begin{aligned}\boldsymbol{\alpha} &= \boldsymbol{e} + 1, \\ \boldsymbol{\alpha} &= \exp(\mathrm{ReLU}(f(x)))\end{aligned}$$

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Subjective Logic

- Strength S of Dirichlet distribution

$$S = \sum_k \alpha_k = \sum_k e_k + 1$$

- Belief mass b_k

$$b_k = \frac{e_k}{S} = \frac{\alpha_k - 1}{S} \text{ and } u = \frac{K}{S}, \text{ subject to } u + \sum_{k=1}^{K} b_k = 1,$$

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minimize the entropy of opinion

$$\underset{\theta}{\text{minimize}}\, H(\mathcal{M}(x)) = -\sum_{k=1}^{K} b_k \log b_k - \lambda u \log u$$

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minimize the entropy of opinion

- First Proposal

$$\underset{\theta}{\text{minimize}}\; H(\mathcal{M}(x)) \;\; \text{subject to}\; |u_\theta(x) - u_{\theta_0}(x)| \leq \delta,$$

Lemma 1. *For any $x \in \mathcal{X}$, we have*

$$\frac{K}{||f(x)||_p + \log K} \leq u \leq \frac{K^{1+1/p}}{||f(x)||_p}, \tag{7}$$

where $f(x)$ is the model output logits, K is the total class number and $||\cdot||_p$ denotes the p-norm.

- Final Proposal

$$\underset{\theta}{\text{minimize}}\; H(\mathcal{M}(x)) \;\; \text{where}\; f(x) = \frac{f(x)}{||f(x)||_p} \cdot ||f(x)||_p^{\text{no\_grad}} \cdot \tau,$$

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EXPERIMENTAL RESULTS

Table 1: Classification accuracy comparison on ImageNet-C (level 5). Substantial (≥ 0.5) improvement and degradation compared to the baseline are highlighted in blue or brown respectively. The detailed results with standard deviation are defer to Table 8 in Appendix C.1.

		Noise			Blur				Weather				Digital			
Methods	COME	Gauss.	Shot	Impul.	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elast.	Pixel	JPEG
No Adapt	✗	35.1	32.2	35.9	31.4	25.3	39.4	31.6	24.5	30.1	54.7	64.5	49.0	34.2	53.2	56.5
PL	✗	49.9	48.8	50.9	48.5	41.2	52.7	42.7	24.6	42.5	63.6	73.1	65.3	44.1	63.9	62.6
T3A	✗	34.6	31.5	35.5	32.7	27.5	40.7	33.5	25.6	30.8	56.5	64.9	50.8	38.0	54.3	58.4
TEA	✗	44.5	39.3	45.8	37.6	35.4	46.4	31.4	8.9	46.4	60.1	72.5	59.5	45.7	62.3	58.9
LAME	✗	34.8	31.9	35.5	30.9	24.4	38.9	30.7	23.4	29.5	53.3	64.2	41.0	32.7	52.8	56.0
FOA	✗	46.6	43.9	48.3	47.1	40.7	49.3	43.7	53.8	52.8	64.2	76.2	63.8	48.5	62.1	64.0
Tent	✗	52.5	52.1	53.4	52.8	47.4	56.7	47.4	10.5	26.4	67.2	74.3	67.3	50.4	66.5	64.6
	✓	53.9	53.9	55.2	55.8	51.8	59.8	52.6	58.4	61.3	71.3	78.2	68.8	57.9	70.5	68.2
	Improve	△1.4	△1.9	△1.8	△2.9	△4.4	△3.2	△5.2	△47.9	△34.9	△4.1	△3.9	△1.5	△7.6	△4.0	△3.6
EATA	✗	56.0	56.1	57.1	54.5	54.8	59.6	58.7	61.8	60.1	71.4	75.3	68.5	62.7	69.0	66.5
	✓	56.1	56.5	57.2	58.0	57.9	62.6	59.3	65.6	63.5	72.6	78.0	69.5	66.6	72.5	70.5
	Improve	△0.1	△0.3	△0.1	△3.4	△3.1	△3.1	△0.6	△3.9	△3.4	△1.3	△2.7	△1.0	△3.9	△3.5	△4.0
SAR	✗	51.9	51.7	52.8	51.5	48.9	55.5	49.5	22.2	46.9	66.2	72.9	65.8	50.9	64.0	62.8
	✓	56.4	56.6	57.4	58.3	56.9	62.9	58.3	65.3	64.5	72.7	78.5	69.6	64.0	71.9	69.7
	Improve	△4.5	△4.9	△4.6	△6.8	△8.1	△7.4	△8.8	△43.1	△17.5	△6.5	△5.5	△3.8	△13.2	△7.9	△6.9
CoTTA	✗	40.3	37.6	41.7	34.3	28.3	44.0	35.6	38.0	43.0	58.8	70.3	58.4	39.8	58.1	59.9
	✓	43.5	40.9	45.5	36.9	29.7	48.1	37.8	40.7	42.0	62.3	73.6	58.9	42.8	63.5	63.8
	Improve	△3.1	△3.4	△3.8	△2.6	△1.4	△4.0	△2.2	△2.7	▽1.0	△3.5	△3.3	△0.5	△3.0	△5.4	△3.9
MEMO	✗	39.7	36.5	39.8	32.4	25.8	40.3	34.7	27.5	32.8	53.5	66.2	56.0	35.7	55.9	58.2
	✓	40.6	37.5	40.6	33.4	26.7	41.2	35.4	28.7	33.7	54.7	67.1	55.9	36.6	57.2	59.3
	Improve	△0.8	△1.0	△0.8	△1.0	△0.9	△1.0	△0.7	△1.2	△0.9	△1.2	△0.8	▽0.1	△0.9	△1.3	△1.1

EXPERIMENTAL RESULTS

Dataset	COME	Tent	EATA	SAR	COTTA
ImageNet-R	✗	37.73 ± 0.03	36.11 ± 0.12	51.90 ± 0.35	36.04 ± 0.06
	✓	39.05 ± 0.05	38.22 ± 0.23	56.40 ± 0.18	37.39 ± 0.08
ImageNet-S	✗	31.63 ± 0.07	36.11 ± 0.19	33.92 ± 0.69	30.84 ± 0.35
	✓	39.22 ± 0.04	39.22 ± 0.32	43.52 ± 0.52	33.82 ± 0.73

EXPERIMENTAL RESULTS

Table 3: Classification and uncertainty estimation comparisons under **open-world** TTA settings, where P^{test} is a mixture of both covariate-shifted samples (Gaussian noise of severity level 3) and a suit of diverse abnormal outliers. Additional results with standard deviation are in Appendix C.2.

		None		NINCO		iNaturist		SSB-Hard		Texture		Places	
Method	COME	Acc↑	FPR↓	Acc↑	FPR↓	Acc↑	FPR↓	Acc↑	FPR↓	Acc↑	FPR↓	Acc↑	FPR↓
No Adapt	✗	64.4	63.7	64.5	69.9	64.4	69.5	64.4	72.5	64.8	65.6	64.3	56.8
PL	✗	69.1	62.8	65.6	71.6	68.8	69.5	68.4	75.9	66.1	66.0	66.6	59.7
T3A	✗	64.4	71.2	64.3	70.0	64.2	75.0	63.7	80.7	64.4	69.0	63.8	69.5
TEA	✗	63.9	63.8	60.5	72.5	62.3	74.6	63.3	79.4	61.0	67.8	61.9	64.5
LAME	✗	64.1	64.4	64.1	72.3	64.1	72.4	64.2	74.0	64.7	68.8	64.0	61.3
FOA	✗	67.8	61.2	66.4	70.5	67.4	66.1	67.1	75.6	65.8	61.4	66.6	54.1
Tent	✗	70.8	63.2	66.2	71.9	69.9	70.7	69.8	77.4	66.4	66.5	68.3	59.9
	✓	72.6	64.7	68.9	64.3	72.5	63.7	72.7	70.7	68.4	60.4	70.7	45.9
	Improve	△1.7	△1.6	△2.7	▽7.6	△2.6	▽7.0	△2.9	▽6.7	△2.1	▽6.1	△2.4	▽14.0
EATA	✗	70.3	63.7	66.4	68.6	70.3	71.5	70.0	77.4	67.3	67.1	68.8	61.9
	✓	73.4	62.7	70.1	60.5	73.2	63.3	73.0	70.5	70.5	55.8	72.3	45.6
	Improve	△3.1	▽1.0	△3.7	▽8.2	△2.9	▽8.2	△3.0	▽6.9	△3.2	▽11.3	△3.6	▽16.3
SAR	✗	69.7	62.3	64.9	71.4	66.9	70.9	67.7	78.1	64.4	64.5	66.1	58.6
	✓	73.1	62.9	69.8	66.3	73.2	65.2	73.5	71.8	69.5	59.4	72.3	49.8
	Improve	△3.5	△0.6	△4.9	▽5.1	△6.3	▽5.6	△5.9	▽6.3	△5.1	▽5.1	△6.3	▽8.8
CoTTA	✗	67.6	63.4	65.3	69.7	70.4	69.5	70.3	76.0	65.8	66.2	66.6	59.2
	✓	70.5	62.5	66.2	68.8	72.4	73.5	72.2	78.7	66.5	64.7	68.9	55.0
	Improve	△2.9	▽0.9	△0.9	▽0.9	△2.0	△4.0	△2.0	△2.7	△0.7	▽1.5	△2.3	▽4.2
MEMO	✗	64.8	69.8	64.8	77.5	64.7	71.9	64.8	77.3	65.0	79.3	64.6	71.5
	✓	65.2	67.8	65.9	76.4	65.2	70.8	65.3	75.0	65.4	76.7	65.3	67.6
	Improve	△0.5	▽1.9	△1.1	▽1.1	△0.5	▽1.1	△0.5	▽2.3	△0.4	▽2.6	△0.7	▽3.9

EXPERIMENTAL RESULTS

Table 4: Classification and uncertainty estimation comparisons under **lifelong** TTA settings. The model is online adapted and the parameters will never be reset, yet the test input distribution might exhibit a continual shift over time.

Methods	COME	Noise			Blur				Weather				Digital			
		Gauss.	Shot	Impul.	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit.	Contr.	Elast.	Pixel	JPEG
No Adapt	✗	35.1	32.2	35.9	31.4	25.3	39.4	31.6	24.5	30.1	54.7	64.5	49.0	34.2	53.2	56.5
PL	✗	49.9	53.4	56.7	46.7	46.1	56.6	51.4	52.6	60.2	68.1	77.7	64.5	51.0	69.0	68.8
FOA	✗	46.5	47.0	51.1	44.8	45.5	52.3	48.1	49.7	57.9	68.0	76.4	63.6	51.7	62.2	65.3
Tent	✗	52.4	56.2	58.7	50.9	51.1	57.7	52.7	54.7	60.5	68.4	77.3	64.6	53.8	69.3	68.6
	✓	54.7	59.0	60.3	51.2	53.7	60.6	57.4	64.1	65.8	71.0	78.6	66.9	62.7	71.8	70.5
	Improve	△2.3	△2.8	△1.6	△0.4	△2.6	△2.9	△4.7	△9.5	△5.3	△2.6	△1.3	△2.3	△9.0	△2.5	△1.9
EATA	✗	55.9	59.5	60.9	56.2	59.2	63.0	61.6	65.7	67.5	72.5	78.6	66.8	67.1	72.2	71.7
	✓	57.9	60.6	61.5	57.0	59.8	63.7	62.3	67.2	68.3	73.7	78.8	69.7	68.6	73.1	71.8
	Improve	△2.0	△1.0	△0.6	△0.8	△0.6	△0.8	△0.7	△1.5	△0.8	△1.2	△0.3	△2.9	△1.4	△0.9	△0.1
SAR	✗	52.0	54.8	56.2	50.2	52.3	56.1	52.8	50.8	26.5	0.1	3.1	0.1	0.1	0.1	0.3
	✓	56.0	60.0	61.1	56.4	58.1	62.9	60.4	66.1	67.4	72.2	78.7	68.0	66.0	72.6	70.9
	Improve	△4.0	△5.3	△4.9	△6.2	△5.8	△6.8	△7.6	△15.3	△41.0	△72.1	△75.6	△67.9	△65.9	△72.5	△70.6
CoTTA	✗	40.3	49.2	57.1	39.8	50.4	55.6	48.3	53.1	61.3	63.9	73.3	62.0	56.5	67.5	66.7
	✓	49.7	61.7	64.2	45.7	57.0	59.0	51.1	58.2	63.1	66.0	73.4	62.9	58.0	68.9	68.2
	Improve	△9.4	△12.4	△7.1	△5.9	△6.6	△3.4	△2.8	△5.1	△1.8	△2.1	△0.1	△1.0	△1.5	△1.3	△1.5

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EXPERIMENTAL RESULTS

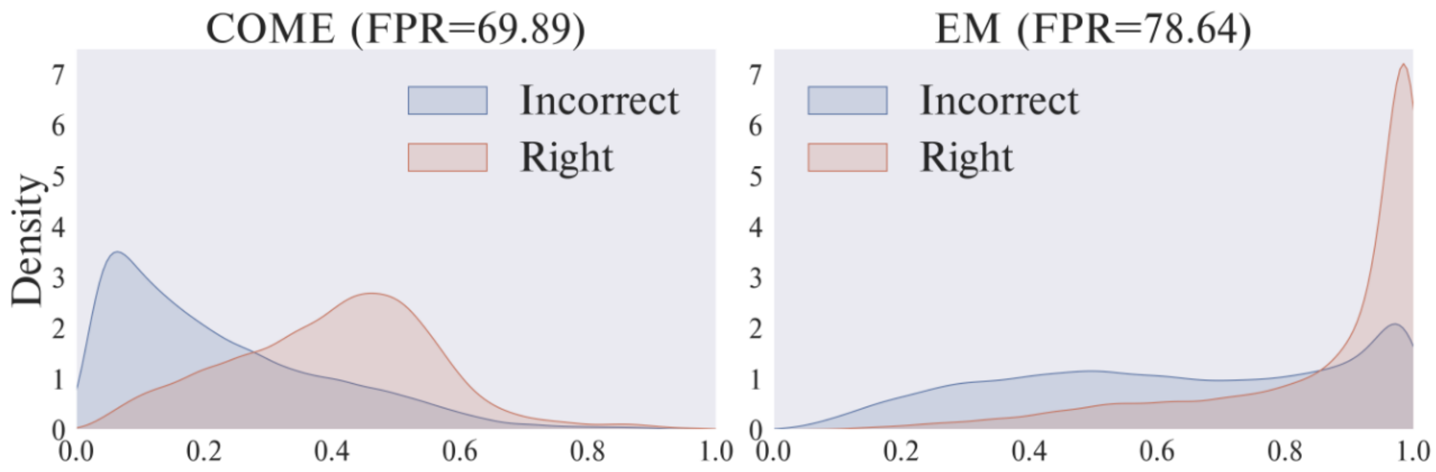

Figure 2: Distribution of model confidence.

		Acc↑		FPR↓	
SL	UC	Mean	Worst	Mean	Worst
✗	✗	52.8	10.6	70.2	95.3
✓	✗	52.7	10.4	70.0	94.9
✗	✓	60.9	25.5	68.0	93.0
✓	✓	61.2	51.7	67.3	71.4

Table 5: Ablation study.

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EXPERIMENTAL RESULTS

Table 6: Additional results with different λ. We report the accuracy when our COME is equipped to Tent on ImageNet-C Gaussian noise level 5 with different λ. The accuracy of the original Tent using entropy minimization is 52.6.

λ	0.10	1.00	10.0	30.0	50.0	80.0	100	150
Acc.	$53.7_{\pm 0.1}$	$53.9_{\pm 0.0}$	$54.6_{\pm 0.1}$	$55.3_{\pm 0.1}$	$55.8_{\pm 0.1}$	$56.2_{\pm 0.1}$	$56.1_{\pm 0.1}$	$10.6_{\pm 0.3}$

Table 7: Additional results with different hyperparameters. We report the accuracy on ImageNet-C Gaussian noise level 5 with different hyperparameters. We implement our COME with Tent. The accuracy of the original Tent using entropy minimization is 52.6.

	$p = 1$	$p = 2$	$p = 3$	$p = \infty$
$\tau = 0.5$	$37.8_{\pm 0.0}$	$38.5_{\pm 0.0}$	$39.1_{\pm 0.0}$	$41.1_{\pm 0.0}$
$\tau = 1.0$	$53.5_{\pm 0.0}$	$53.8_{\pm 0.1}$	$53.2_{\pm 0.0}$	$47.2_{\pm 0.1}$
$\tau = 1.2$	$54.7_{\pm 0.0}$	$54.7_{\pm 0.1}$	$54.0_{\pm 0.1}$	$48.9_{\pm 0.0}$
$\tau = 1.5$	$54.2_{\pm 0.2}$	$54.2_{\pm 0.1}$	$53.2_{\pm 0.1}$	$49.1_{\pm 0.1}$


Thank You!